

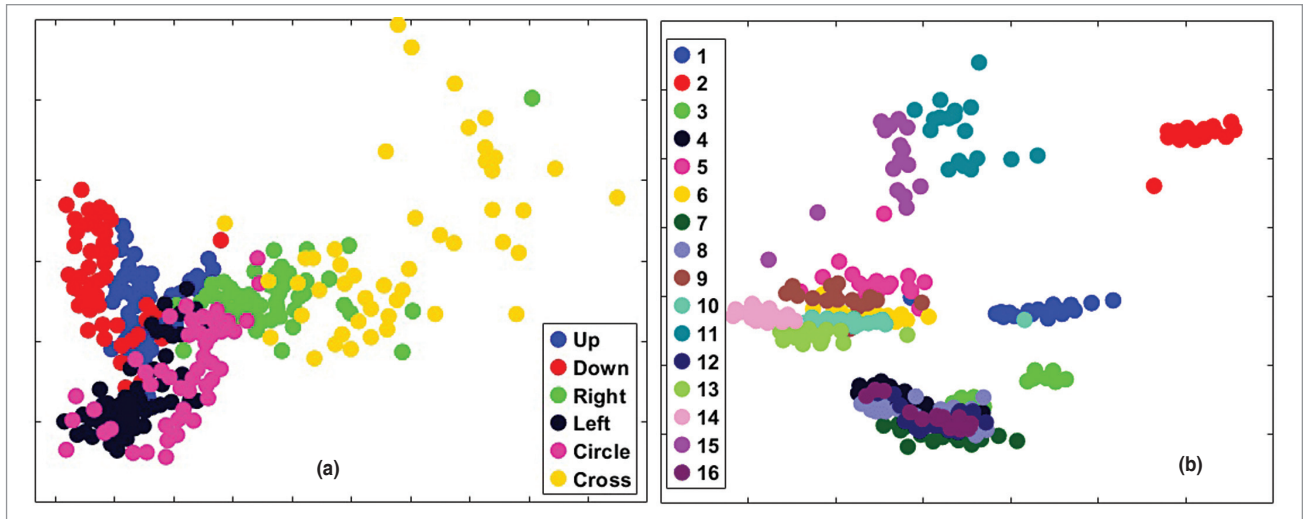


Fig. 4: t-SNE of (a) joint activity recognition, and (b) localisation

smaller and more efficient model. As the number of parameters decreases, the FLOPs reduce as well, reducing the computational complexity of the model. The weights are removed based on a preset criterion like l_1 or l_2 -norm or their contribution to the loss function. Weights with lesser magnitudes are considered less significant and thus removed. The amount of parameters to be removed can be controlled by controlling the sparsity introduced in the model while pruning. As the parameters are removed, although less significant, the model performance degrades, and an optimal balance between size and accuracy is needed. The methodology used is shown in Algorithm 1.

Algorithm 1 CSI-based joint activity recognition and localisation
Input: CSI Data $X \in \mathbb{R}^{N \times K \times 2}$

Output: Predicted Activity $\hat{A}$ and Location $\hat{L}$

Step 1: CSI data collection

1. Collect raw CSI data from Wi-Fi signals in the environment.
2. Pre-process the CSI data by removing noise and normalising.

Step 2: Model training (deep learning)

3. Train a convolutional neural network (CNN) using labelled CSI data.

4. Optimise model weights θ using backpropagation to minimise the the loss function.

Step 3: Model compression

5. Pruning: Remove insignificant weights to reduce model size.
6. Quantisation: Convert model weights to lower precision.

Step 4: Optimisation and evaluation

7. Measure accuracy and latency before and after compression.
8. If performance degrades significantly, adjust compression parameters and retrain.

9. End if

Step 5: Deployment on Edge devices

10. Deploy the compressed model on an edge device (e.g., Raspberry Pi 4).
11. Perform inference on real-time CSI data to predict $\hat{A}$ and $\hat{L}$.

Performance evaluation

The model can be compressed significantly by combining pruning and quantisation. First, pruning reduces the model's parameter count, and then quantisation further reduces the model size. The training process is treated as a classification task.

Fig. 3 illustrates the effect of compression on accuracy. The

original model, without pruning or quantisation, achieves 99.25% accuracy for localisation and 95.32% for activity recognition.

When compressed to 80% sparsity with 7-bit quantisation, an optimal balance is achieved between size, complexity, and performance. Under these conditions, localisation accuracy is 84.89% and activity recognition accuracy is 66.55%. The t-SNE (t-distributed Stochastic Neighbour Embedding) visualisation is shown in Fig. 4.

With reduced model size and complexity, inference time is also significantly improved. When deployed on a Raspberry Pi 4, the compressed model achieves an inference speed of 10.16ms, compared to 49.93ms for the uncompressed model—a nearly 80% reduction.

Thus, model compression enables efficient deployment of CSI-based models on edge devices.

CSI-based tracking holds promise for applications in smart cities, retail environments, and industrial automation. Future developments may focus on improving model efficiency, integrating with emerging wireless standards such as 6G, and enhancing privacy via federated learning. **EFY**